

Team 2:- Theme 5:- Transaction based

Inventory Management system

8/9/26 - Theme selection... "3:15 pm"

9/9/26 - Problem statement proposal to KMMR
"12:15 pm" Sir,

[From the term Inventory Management system we decided to brainstorm on Warehouses for managing there Inventory Stakeholders:- Warehouse Manager !!!

Warehouse staffs

Logistic personell.

Clients:- Blinkit, Zento, Instamart
[Quick Commerce Pvt]

11/9/26 - Detailed ps explanation with As-is workflow,
"12:15 pm" To-Be Vision, Inscope & outscope, Assumptions
Constraints, Evidence, Feasibility study,
High level Project Goals etc.

At 4:00 pm we got official doc from
HOD CSE & we followed that format
& updated our Git with 30 documents.

Remarks:- PS is upto the mark, ideate
more...

Make a notebook & update every
tiny details, start working on
KPI Indexing & Engines.

15-9-26 - Detailed Review & Judgement of Doc.
12:15
Remarks:- More focus on core subject.
• Use Technical terms while Explaining.

22-9-26 - Gatu - O Evaluation with HOD - CSE,
1:30 - 4:30 KMMR, SB, VS, SSK.

Remarks:- make your PS / solution more complex.

• Discuss KPI.

• Scenario based discussion of different users.

• Include customers into PS.

• Create complex query for fetching / updating data by joining multiple tables.

25-9-26 - 2 Complex Scenarios having 10 Use Cases was presented.

12:15.

• Done with discussion of KPI (didn't show to sir)

• Included customers as Primary stakeholders & our solution is to satisfy the customer using Quick Commerce platforms.

• Explained our confusion regarding clients if we consider multiple clients

Should we have them in
Single warehouse

Each client having
different warehouse

Which is feasible & beneficial?

Remarks:- Note down o HDLC,
o FR o VFR o Use Cases o Vlan stores

Basically SI, still
we must show this on 26-9-26
Meeting arranged with KMMR
(mentor) sir on 26-9-26, 12:15.

Remarks:- Note down o HDLC,
o FR o NFR o Use Cases o Van stores

Basically SI, still
we must show this on 26-9-26
Meeting arranged with KMMR
(mentor) sir on 26-9-26, 12:15.

28-9-26 - Explained all Failure Cases & their solut^{ns}
using concepts.

Showed KPI's research.

Remarks:- Sir is convinced &
satisfied with complexity &
feasibility of our ps. (80).

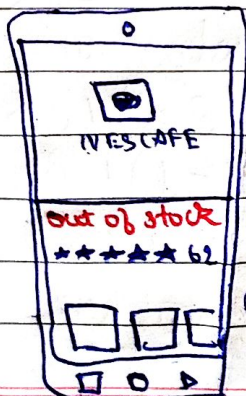
29-9-26 - GO Evaluatⁿ 2.

Our P3/Solut² Proposal.

Core Problem:-

Multiwarehouse inventory imbalance causes localized stockouts despite network-level stock availability. The core challenge is to dynamically detect shortages, identify an eligible source warehouse, calculate safe transfer quantity & execute redistribution² while maintaining transaction consistency, concurrency control & inventory traceability.

- 1) Multiwarehouse inventory
- 2) Demand-driven shortage detection.
- 3) Source selection.
- 4) Safety stock.
- 5) Transfer quantity calculation.
- 6) Concurrent requests.
- 7) Inventory reservation.
- 8) Approval workflow.
- 9) Logistic assignment
- 10) Transfer lifecycle.
- 11) Inventory reconciliation.
- 12) Audit Trail.



“Customers encounter OUT OF STOCK products even though the same inventory may exist in nearby warehouses, resulting in lost sales and poor customer satisfaction.

The absence of intelligent real-time stock re-distribution mechanism causes localized shortages while available inventory remains underutilized elsewhere.

Situations (S)

S1:- Normal stock ... stock monitoring.

SWD has 100 cans.

Customer buy 10.

$\therefore$ stock = 90, No redistributed?

Orders are coming continuously.

DBM S: stores inventory for each warehouse - in

Inventory table.

SE: Defines the inventory monitoring module.

WebT: Displays live inventory on manager dashboard.

Separate inventory by Warehouse.

each warehouse needs independent stock records.

SI This was just an intro.

Core Scenario:-

Assume:-

Warehouse	Locat ⁿ	Current Stock	Safety Stock	Avg Daily demand
DWD	Dharwad	18	20	35
H-1	Hubli	40	40	18
H-2	Hubli	65	25	30
B-1	Belgaum	210	80	50
B-2	Belgaum	55	20	15
HVR	Haveri	35	15	12

Situation:- DWD Experiencing a sales spike.

Hour	Sales
9 am	5
11 am	7
1 pm	10
3 pm	13

∴ Current stock becomes 18.

DSA:- Instead of $18 < 30$, system calculates.

$$\begin{aligned}\text{Days of Inventory} &= \frac{\text{Current Stock}}{\text{Average Daily Demand}} \\ &= \frac{18}{35} = 0.51 \text{ days}\end{aligned}$$

Meaning:- Stock will likely finish in next 12 hrs.

DBMS:-

- Fetches Current Stock
- Historical Sales
- Safety Stock

→ Follows Event Driven Architecture.

SE:- Shortage Detection Module automatically raises a stock request.
Order Service → Inventory Service → Inventory Changed Event

WebT:- Critical — Predicted stock exhaustion within 12 hrs (Dashboard display).

→ SQL Transaction:- A customer buys 3 cars at DWD
Begin;

update INVENTORY
SET available_qty = available_qty - 3,
updated_at = **NOW()**
Where warehouse_id = 3.
AND product_id = '20001';

COMMIT;

Inventory table:-

	inventory_id	warehouse_id	product_id	available_qty
Before	301	3	20001	21
After	301	3	20001	18

reserved_qty Safety Stock
0 20
0 20

Inventory
change
event

Now the Question why 35 as average

suppose past week sales.

day

1

32

2

35

3

37

4

34

5

36

6

38

7

33

$$\text{Average} = 35$$

Total
7

Also.

Trend Analysis.

Algorithm compared

3 day average

MA_3

7 day average

MA_7

$$\text{if } MA_3 > MA_7$$

Demand is rising.

called Trend detect?

Warehouse classification?

$$DOI > 3$$

$$DOI (1-3)$$

$$DOI < 1$$

status

Healthy

Low

Critical

Days of Inventory.

Request Generation:

SE:- Shortage detect module creates a STOCK_REQUEST.

DBM:-

INSERT INTO STOCK REQUESTS (

destination - warehouse - id,

created by,

priority,

requested - status,

requested - at

)

A/o Manager

manually creates it

VALUES (

3,

SYSTEM,

'CRITICAL',

'PENDING',

The system creates it.

WebT

Frontend calls: GET /inventory /critical.

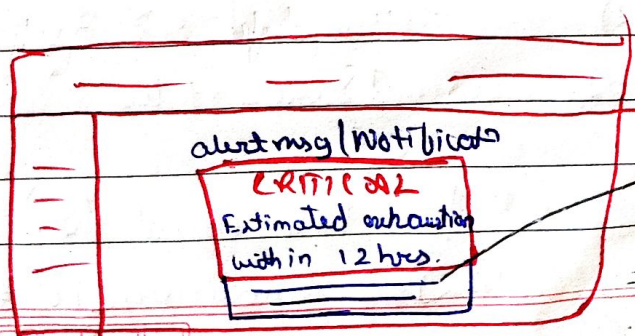
Returns: "warehouse": "Dharamad",

"product": "Diet coke",

"stock": 18,

"DOI": 0.5,

"Priority": "critical".



must think what to add?

=> something like approve request generated by our system, to request

INTERNAL FLOW DIAGRAM

Customer Order

Inventory Transaction

Inventory Changed Event

Detection Module ——— DDI + Trend Analysis ——— (critical AP)

Stock Request

Priority

Classification

Manager

Dashboard

Space for Monitor Inputs

Scenario 2:- Intelligent Source Warehouse selection.

After S1, DWD generated Critical Stock Request.

Our System must answer: Which warehouse should supply the stock?

Not the nearest, but optimal warehouse.

S1:- DBMS.

Redistribution module queries all warehouses
(except destinst?).
→ writes Query

SELECT

warehouse_id,

" name

available_qty,

safety_stock,

warehouse_status

FROM INVENTORY

JOIN WAREHOUSES USING (warehouse_id)

WHERE product_id = 'D1001'

AND warehouse_id <> 3

AND warehouse_status = 'ACTIVE';

→ <> is != (not equal to) because
DWD's warehouse_id = 3.

Result.

Warehouse	Avl	Safety	Transferable
H-1	140	40	100
H-2	65	25	40
B-1	210	80	130
B-2	55	20	35
HVR	35	15	20

Filters inactive warehouses.

Calculates Transferable stock (Avl - safety)

Provides clear input for algorithm.

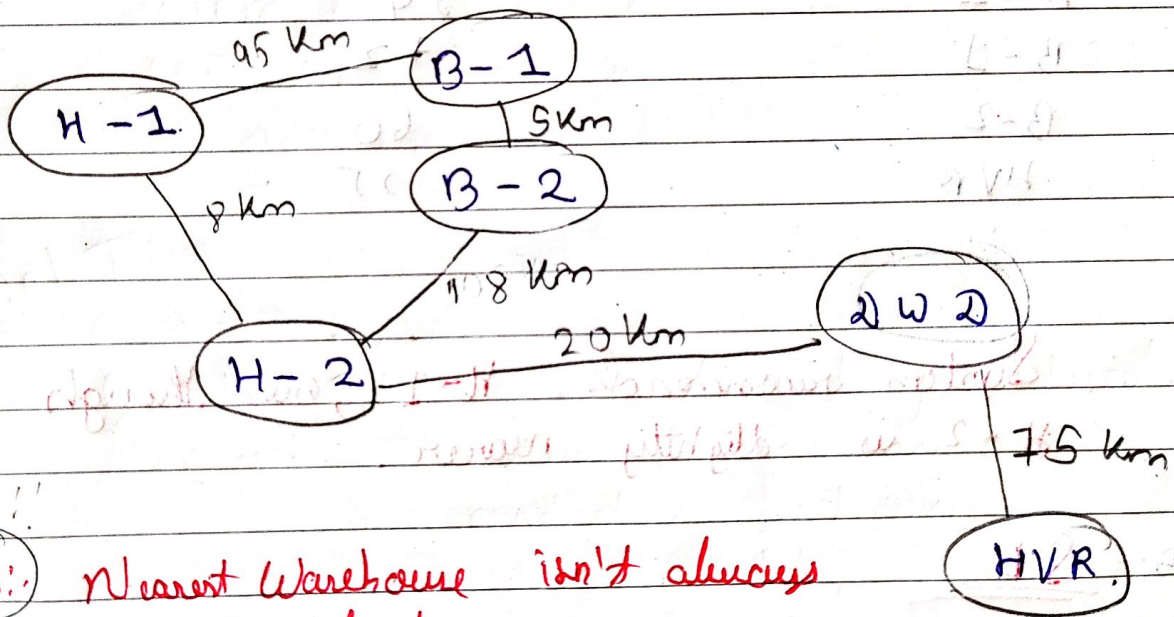
Why?

S2:

Dijkstra

→ guarantees min-cost route in weighted graph.

Dijkstra's Algo finds shortest path



S3:

Nearest Warehouse isn't always the best.

assume 2W2 needs 60 but H-2 has 40
 ∴ requirement not satisfied

we can see H-1 has 100 Transferable
 but 28 km distance

↳ +8 increase in distance.

A difference in 8 km allows single delivery rather than 2 deliveries (40 + 20)

from somewhere else.

34) Highlighted Warehouse Ranking (DSS).

Distance (D) 350/0
 Transferable stock (S) 350/0
 Future Demand (F) 200/0
 Warehouse status (W) 100/0

Formula

$$\text{Score} = 0.35 (D) + 0.35 (S) + 0.20 (F) + 0.10 (W)$$

Calculated scores

W	Score
H-1	90
H-2	84
B-1	72
B-2	60
HVR	55

System recommends H-1, even though H-2 is slightly nearer.

35: SE

Redistributed 2 Module now creates recommended object.

same: 200
 distinct: H B 2
 P-id: 20001
 distance: 28
 using 90

SB:- Udile Dashboard

API :- GET / recommendation / 165

Recommended Source

Hubli - 1

Best
match

distance = 28km Score 90 Qty: 60 cars

28km

90

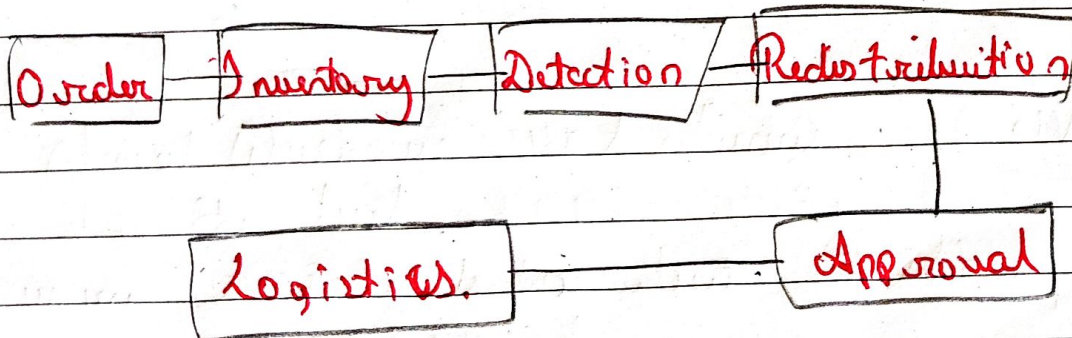
60

Reject

Approve

In SE what exactly is Module?

In SE concept called Modular Architecture
where each part is responsible for 1 functionality.



Use Case: - Interact b/w Actor (User/System) and the software to achieve one specific goal.

1. Intelligent Shortage Detect ☺

Actor System

Goal Detect a warehouse shortage before stock-out.

Trigger Inventory changes after a customer order.

Main Flow

1. Update Inventory after sales.
2. Retrieve current stock, safety stock, & historical sales.

3. Calculate moving average demand

4. Calculate DOI.

5. Classify stock as Healthy, Low, or critical.

6. Create stock request automatically if critical.

Core Subjects: DSA, DBMS, IF, WebT.

2.

Scenario 3: Concurrent stock Request Conflict (Race Condition)

Situation H-1 has 100 Transferable Cars.

At 2:00 pm, 2 warehouses send request simultaneously

DW2 60 required

HVR 50 " "

without protect, both request read 100 & allocate 110 cars, creating -ve inventory.

S1: DBMS detects Concurrent Access.

Both request start transactions.

1. BEGIN;

First transaction locks inventory row.

```
SELECT available-qty.  
FROM INVENTORY  
WHERE warehouse-id = 1  
FOR UPDATE;
```

S2: First transaction commits.

Inventory becomes $100 - 60 = 40$.

```
UPDATE INVENTORY  
SET reserved-qty = reserved-qty + 60  
WHERE warehouse-id = 1;
```

33: Second Transactⁿ -> Rechecks.

HVR now reads out = 40 which is $40 < 50$.

Recheck Unit = Engine Recalculates.

Possible outcomes

- o reduce quantity.
- o choose another warehouse
- o or keep request pending.

2BM $\rightarrow$ Row level locking.
"Concurrency Control"

Scenario 4: Source Warehouse Demand suddenly increases.

Situation: before source is transferring 60 cars
After a dispatch receives 45 customers order
Current Inventory changes rapidly

$\nearrow$ with remaining
10-15 or
might be 0.

Wrong Approach $\rightarrow$ Cancelling the transfer.
causes delivery delay.
double logistics cost,
customer dissatisfaction.

Solutⁿ :- Inventory overcommit? For ex:
If there are 100 cars: Reserve 40 cars
& Max Transferring of cars must be set
Ex: 60 cars.

In this way we won't be able to face this issue.

Scenario 5: No warehouse can fully satisfy the Request.

Ex: If DVD needs 150 cars, but including all the existing warehouses if we can't fulfill the request then system uses **Greedy Allocation strategy** (DSA)

where allocated ≤ 150

→ **Partial Fulfillment.**

GA's Explanation:-

Dijkstra finds shortest route from one warehouse to another. Greedy is used when one warehouse alone cannot fulfill the demand, it allocates stock from the best warehouse first, then next best, until the required quantity is completed.

Scenario 6: Network Failure During Approval

When Manager clicks Approve, **Internet disconnects**

Transaction begins & when conn^t is lost,

suppose request² succeeds,

Transaction creat² down 't.

without protect², database becomes inconsistent.

Solutⁿ: Rollback: (DBMS)

Inventory remains Unchanged.

Atomicity in ACID



Either everything succeeds or nothing changes.

libT



Retry msg.

My Questⁿ ???

Should we consider Multiple Clients?

If Yes :- Should we have separate warehouse for each clients

Else

1 Warehouse, multiple clients.

Else

Single client.

Ans: Separate warehouse networks for each client - commerce platforms.

7 Blinkind Zpto maintains separate inventory records.

Problems with Collaborating - Inventory Ownership

Billing - Each platform has separate invoice, commissions.

Operatⁿs - Different packing, picking

Security: One company shouldn't access another company's inventory.

We can create/design database as multi-tenant.

Id	platform-id	Platform
1		Blinkit
2		Zepto
3		Instamart

Every warehouse belong to a platform.

Ex:

warehouse id	platform id	Warehouse
BH-1	1	Blinkit Hubli-1
BH-2	1	" " - 2
ZH-1	2	Zepto Hubli
IH-1	3	Instamart Hubli

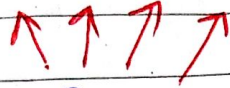
Geography

Tier - 1. — Karnataka divisions.

Belgaum	Gulbarga
Mysore	Bangalore

Mother warehouse - 1/2
dark warehouse - n.

Redistributⁿ works
only inside the Divisions.



Tier - 2, Regional / State cities, India.

Bangalore Regional Hub.
Telangana Regional "
Pune " "

Rajasthan Regional Hub.

Belgaum
Gulbarga
Mysore
Bangalore.

→ If Belgaum / Gul / Mys / Nearby warehouse doesn't have the stock then we Regional, request its regional distributⁿ centre (Bangalore).

Until here Automated.

Tier 3 - Rajasthan as Ex.

Only if Regional inventory is also insufficient does the system generate a long-distance replenishment request?

It's no longer Redistributⁿ — its strategic Replenishment.

For more guests:- Should we design ^{different} V2 for every other "Tiers" ???

different

every other Tiers --- ???? ?